Section 2.7 — Function Algebra — Answer Key

Math 103: College Algebra

Combined functions

Define the following functions:

$$f = \{(-7,-9), (-1,-3), (1,\ \ 3), (5,\ \ 0), (9,-8)\}$$
$$g = \{(-7,-4), (\ \ 1,-8), (5,\ \ 1), (7,\ \ 0), (9,-7)\}$$
$$h = \{(-7,\ \ 0), (-1,-5), (1,-7), (7,-1), (9,-5)\}$$

Find each of the following. If the given value is not in the domain of the given function, write "undefined."

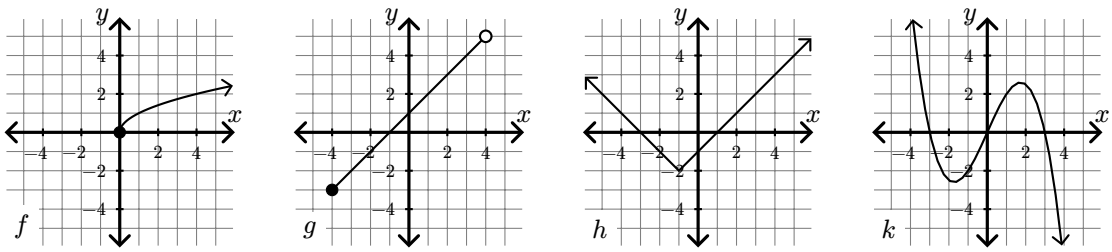
1. 56
2. undefined
3. -1
4. 0
5. -13
6. -11
7. undefined
8. 0
9. undefined
10. -1
11. $\frac{5}{3}$
12. -8

Find each domain.

13. $\operatorname{Dom} f = \{-7, -1, 1, 5, 9\}$
14. $\operatorname{Dom} g = \{-7, 1, 5, 7, 9\}$
15. $\operatorname{Dom} h = \{-7, -1, 1, 7, 9\}$
16. $\operatorname{Dom}(g + h) = \{-7, 1, 7, 9\}$
17. $\operatorname{Dom}(h - f) = \{-7, -1, 1, 9\}$
18. $\operatorname{Dom}(f \cdot g) = \{-7, 1, 5, 9\}$
19. $\operatorname{Dom}\left(\dfrac{h}{g}\right) = \{-7, 1, 9\}$

Combined Functions from Graphs

Consider the functions graphed below.



Find each of the following. If the given value is not in the domain of the given function, write "undefined."

20. -3
21. 2
22. 0
23. 0
24. undefined
25. undefined
26. 0
27. undefined

Find each domain.

28. $\operatorname{Dom} f = [1, \infty)$
29. $\operatorname{Dom} g = [-4, 4)$
30. $\operatorname{Dom} h = \mathbb{R} = (-\infty, \infty)$
31. $\operatorname{Dom} k = \mathbb{R} = (-\infty, \infty)$
32. $\operatorname{Dom}(f - g) = [0, 4)$
33. $\operatorname{Dom}(k + f) = [0, \infty)$
34. $\operatorname{Dom}(h \cdot k) = \mathbb{R} = (-\infty, \infty)$
35. $\operatorname{Dom}\left(\dfrac{g}{k}\right) = [-4, -3) \cup (-3, 0) \cup (0, 3) \cup (3, 4)$

Combined Functions

$f(x) = 3x^2 - 8x + 4$ $\quad g(x) = x^3 + 7x^2 - 4x - 28$ $\quad h(x) = x - 2$ $\quad k(x) = 3x^2 - 19x - 14$

Use the functions above to find the following.

36. -40
37. 0
38. 0
39. undefined

Write expressions for each of the following. Simplify your answer as much as possible.

40. $(f - k)(x) = 11x + 18$
41. $(g + f)(x) = x^3 + 10x^2 - 12x - 24$
42. $(f \cdot h)(x) = 3x^3 - 14x^2 + 20x - 8$
43. $(h - g)(x) = x^3 + 7x^2 - 5x - 26$
44. $(h + k)(x) = 3x^2 - 18x - 12$
45. $(f \cdot k)(x) = 9x^4 - 81x^3 + 122x^2 + 36x - 56$

Write expressions for each of the following. Do not simplify your answer.

46. $\left(\frac{f}{h}\right)(x) = \dfrac{3x^2 - 8x + 4}{x - 2}$
47. $\left(\frac{g}{h}\right)(x) = \dfrac{x^3 + 7x^2 - 4x - 28}{x - 2}$

Find each domain.

48. $\operatorname{Dom} f = \mathbb{R} = (-\infty, \infty)$
49. $\operatorname{Dom} g = \mathbb{R} = (-\infty, \infty)$
50. $\operatorname{Dom} h = \mathbb{R} = (-\infty, \infty)$
51. $\operatorname{Dom} k = \mathbb{R} = (-\infty, \infty)$
52. $\operatorname{Dom}(f + g) = \mathbb{R} = (-\infty, \infty)$
53. $\operatorname{Dom}(h - k) = \mathbb{R} = (-\infty, \infty)$
54. $\operatorname{Dom}(f \cdot h) = \mathbb{R} = (-\infty, \infty)$
55. $\operatorname{Dom}\left(\frac{g}{k}\right) = \{x \mid x \neq -\frac{2}{3}, 7\} = \left(-\infty, -\frac{2}{3}\right) \cup \left(-\frac{2}{3}, 7\right) \cup (7, \infty)$